

How Much of BERT Does Text Classification Need?

A controlled comparison with DistilBERT

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THE QUESTION AND THE SETUP

PROBLEM

DistilBERT claims 97% of BERT's GLUE score with half the layers. We tested that on three datasets.

Models	bert-base-uncased (12 layers, 109.5 M) and distilbert-base-uncased (6 layers, 67.0 M)
Data	SST-2 (872 test sentences), AG News (7,600), Yelp Polarity (38,000, train cut to 100 k)
Recipe	AdamW at 2×10^{-5} , 10% warm-up, 2 epochs, batch 32, fp16, seed 42
Selection	best validation macro-F1, then one pass over test
Runs	24 fine-tunings plus a latency benchmark, all on one RTX A6000

REMINDER

Only the model changes between two columns of any table. Training loop, splits, seed and GPU stay the same for both.

BASE RUNS

Dataset	Model	Test (%)				Efficiency			
		Acc	Prec	Rec	F1	Params (M)	Latency (ms)	Memory (MB)	Train (min)
SST-2	BERT	92.32	92.32	92.31	92.31	109.5	5.56	430	4.2
SST-2	DistilBERT	91.28	91.34	91.25	91.27	67.0	2.60	269	2.7
AG News	BERT	94.41	94.43	94.41	94.41	109.5	5.61	433	9.1
AG News	DistilBERT	94.50	94.52	94.50	94.50	67.0	2.61	272	5.5
Yelp	BERT	96.52	96.52	96.52	96.52	109.5	5.11	436	16.0
Yelp	DistilBERT	96.06	96.06	96.06	96.06	67.0	2.74	275	8.2

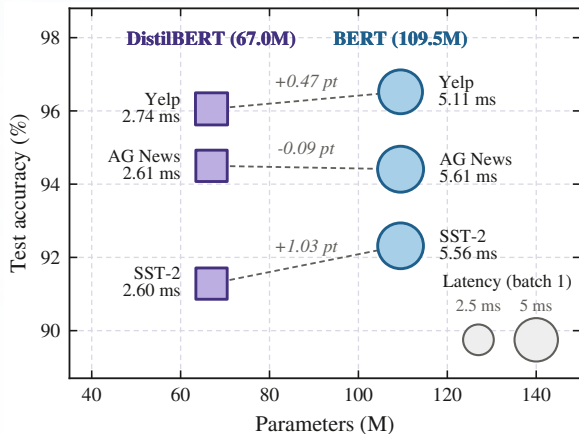
EXAMPLE

BERT leads SST-2 by 1.04 points and Yelp by 0.46. On AG News DistilBERT is ahead by 0.09, which is seven test items out of 7,600. We call that a tie.

REMINDER

DistilBERT scores 91.28% on SST-2. Sanh et al. report 91.3, so our splits and recipe reproduce the published number.

SIZE, ACCURACY AND LATENCY

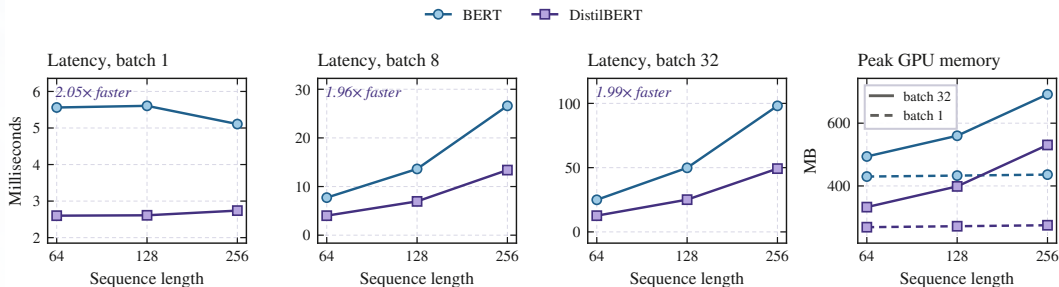


DistilBERT sits 42.5 M parameters to the left of BERT, with bubbles half the size. The vertical distance inside each pair is what the smaller model costs in accuracy.

EXAMPLE

With 61% of the parameters, DistilBERT keeps 98.9% of BERT's accuracy on SST-2 and 99.5% on Yelp. On AG News it keeps all of it.

CONTROLLED LATENCY BENCHMARK



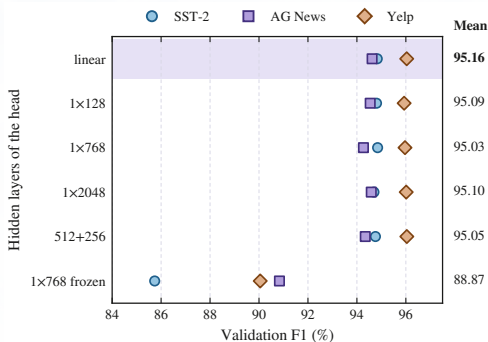
EXAMPLE

DistilBERT wins all nine grid points, by 1.86× to 2.15×. At batch 32 and 64 tokens it handles 2,535 samples per second against 1,282. Its peak memory is 62% of BERT's at batch 1 and 77% at batch 32 with 256 tokens.

CHALLENGE

The timings logged after each run put BERT ahead on SST-2 (3.16 against 3.50 ms). Twice the layers can't be faster. We threw those out and timed both models again in one process, in fp32, with 20 warm-up passes and 50 synchronized ones.

SIX CLASSIFIER HEADS ON DISTILBERT



Every head reads the [CLS] state. Width goes from 128 to 2048, depth from zero to two hidden layers, and one run freezes the encoder.

EXAMPLE

With a trainable encoder the five heads land between 95.03 and 95.16 mean validation F1. Their order changes from dataset to dataset.

PROBLEM

Freezing drops the mean to 88.87. With the same head it loses 9.1 points on SST-2, 5.9 on Yelp and 3.4 on AG News.

PROPOSAL

The linear head wins on the mean. It is also the smallest and trains fastest.

THE LINEAR HEAD ON BOTH MODELS

Dataset	Model	Performance (%)					Efficiency			
		Acc	Prec	Rec	F1	Val F1	Params (M)	Latency (ms)	Memory (MB)	Train (min)
SST-2	BERT	92.89	92.94	92.86	92.88	95.41	109.5	5.56	430	4.2
SST-2	DistilBERT	90.94	90.99	90.91	90.93	94.83	66.4	2.60	269	1.7
AG News	BERT	94.53	94.56	94.53	94.53	94.81	109.5	5.61	433	9.0
AG News	DistilBERT	94.42	94.43	94.42	94.42	94.62	66.4	2.61	272	5.4
Yelp	BERT	96.53	96.53	96.53	96.53	96.61	109.5	5.11	436	16.0
Yelp	DistilBERT	96.10	96.10	96.10	96.10	96.03	66.4	2.74	275	8.2

EXAMPLE

BERT takes all three datasets: 1.95 points on SST-2, 0.43 on Yelp and 0.11 on AG News.

COMMENT

Without its tanh pooler, BERT climbs to 92.89 on SST-2 from 92.32. DistilBERT slips from 91.28 to 90.94, so the gap opens from both ends.

WHY THE NUMBERS LOOK LIKE THIS

DEFINITION

Task. AG News labels hang on words like earnings or playoffs, and six layers already see them. Both models trip on the same stories. BERT sends 140 Business items to Sci/Tech and 90 back, DistilBERT 140 and 92. SST-2 sentences turn on negation and contrast, where depth pays.

DEFINITION

Head. Fine-tuning bends the [CLS] vector until one linear layer separates the classes.

DEFINITION

Latency. At batch 1, BERT barely moves with length (5.56, 5.61 and 5.11 ms). The GPU waits on kernel launches, and DistilBERT has half as many. From batch 8 up, time grows with input length and DistilBERT does half the arithmetic.

COMMENT

Memory closes in as the batch grows. Activations add about the same amount to both models.

CONCLUSION

- DistilBERT is our default for these tasks. It keeps 98.9% to 100% of BERT's accuracy with 61% of the parameters, two thirds of the memory and half the latency.
- BERT's margin has a price. On AG News, 0.11 points will rarely justify doubling the inference bill. On SST-2, 1.95 might.
- One linear layer on [CLS] is enough. Freezing the encoder is the choice that hurts.

PROBLEM

Limitations. Every run used seed 42, so we have no intervals and gaps of a few tenths count as ties. SST-2 has 872 test sentences, and 1.04 points is nine of them. Two epochs is the low end of what BERT's authors recommend.

THANK YOU

Questions?

